Lakshay Chauhan

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Education

Indraprastha Institute of Information Technology (IIIT) Delhi — Jul 2021 – Jun 2025
Bachelor of Technology in Computer Science and Engineering — *Delhi, India*

Work Experience

Member of Technical Staff — Jul 2025 – Present
*ZL Technologies, Inc | **Website*** — *Hyderabad, India*

- Developed task synchronization and bulk parallel processing mechanisms across clusters within ZL's analytics module, achieving over 100% performance improvement.
- Implemented task completion time estimation and duplicate item detection for large-scale data loading workflows processing millions of items.

Research Assistant | *Gitlab Project Organization* — May 2024 – May 2025
*Networks and Systems Security Lab - IIITD | **Website*** — *Delhi, India*

- Researched mechanisms to mitigate video streaming quality degradation during fallback from QUIC to TCP caused by UDP blocking.
- Implemented QUIC-based notification signaling and kernel congestion window state reuse to bypass TCP slow start, reducing buffering and improving user experience during fallback transitions.

Projects

Quorum — Distributed Key-Value Store | *Repository* — Jan 2026 – Mar 2026

- Engineered a distributed key-value store in Java, focusing on scalable backend architecture, inter-node communication, and fault-tolerant request processing.
- Designed data replication and synchronization mechanisms to maintain consistency across multiple nodes under concurrent workloads. Built modular networking, persistence, and concurrency components enabling reliable request handling and distributed systems experimentation.

Quill — Task-Based Parallel Runtime Library | *Repository* — Sep 2025 – Dec 2025

- Developed a task-based parallel runtime library in C++ for efficient execution of concurrent workloads across worker threads.
- Implemented work scheduling, task dispatch, and synchronization primitives to maximize throughput while minimizing coordination overhead.
- Designed reusable abstractions for parallel execution, improving scalability and resource utilization in multithreaded applications.

Analysis of Congestion Control algos in TCP Variants | *Repository* — Aug 2024 – Nov 2024

- Conducted performance analysis of TCP congestion control algorithms using C++, evaluating throughput, latency, and network behavior under varying conditions.
- Built benchmarking and data collection workflows to compare algorithmic trade-offs across multiple TCP variants. Investigated protocol-level optimization techniques and system bottlenecks, strengthening understanding of low-latency networking and performance engineering.

Technical Skills

Languages: C, C++, Rust, Assembly, Java, Kotlin, JavaScript, Python, Bash, Haskell
Frameworks: PyTorch, Django, ReactJS, Numpy, Pandas, Tauri, LibGDX, Raylib
Developer Tools: Emacs, Linux, Git, GDB, Markdown, Google Cloud Platform, OpenLiteSpeed, SqlLite3
Technical Electives: Data Structures & Algorithms, Operating Systems, Cryptography, Database Management, Computer Security, Computer Networks, Compilers, Machine Learning, Natural Language Processing

Awards

- Awarded the **Summer Undergraduate Research Fellowship** (2023) by **IRD-IIITD** for the project *"Utilizing Ultrasonic Distance Sensors as a Mapping Tool to Design User-Friendly CST"*.
- Awarded the **CHANAKYA Fellowship** (2024) by **iHub Anubhuti Foundation** for the project *"A Unified Approach to User Emotion Detection through Emojis and Textual Analysis"*.